

João Marcos Amario de Sousa

Ph.D. Candidate in Genetics and Plant Breeding

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ABOUT ME

Ph.D. candidate in Genetics and Plant Breeding at the Federal University of Viçosa (UFV), advised by Prof. Kaio Olimpio das Graças Dias, focusing on genomic prediction and the integration of genomic and environmental information to model genotypes × environments interaction and predict genotype performance in untested environments. I am currently a visiting scholar (sandwich Ph.D.) at the University of Illinois Urbana-Champaign, advised by Prof. Alexander Lipka, developing statistical and computational models applied to plant breeding. My work spans quantitative genetics, mixed models, environmental covariates, multi-environment trial analysis, spatial modeling and the development of analytical tools in R, aiming to deliver robust solutions that breeding programs can put into practice.

EXPERIENCE

Ph.D. in Genetics and Plant Breeding

Idata, Federal University of Viçosa · 2024 – present

- Identify methodological gaps in plant breeding research and design analytical strategies that combine mixed models and statistical learning to support selection decisions.
- Apply mixed models in ASReml to predict genotype performance in untested environments, integrating environmental covariates and factor analytic models for cultivar recommendation.
- Develop and maintain *qgtools*, an open-source collection of R functions I use daily for two-stage multi-environment analysis, environmental covariate assembly and spatial genomic and environmental prediction.
- Maintain public repositories on GitHub, using version control and reproducible coding practices.

M.Sc. in Genetics and Plant Breeding

Federal University of Viçosa · 2022 – 2024

- Investigated statistical genetics models with residual and genetic covariance structures to select tropical wheat populations.
- Conducted tropical wheat field trials, from planting to harvest and data collection, building a solid foundation in field experimentation that now underpins the analyses and models I develop in my Ph.D.

Agricultural Consultant

Fazenda Mundo Novo, Darcinópolis, TO, Brazil · 2022 – 2026

- Provided remote technical advisory on maize production and pasture management, with recommendations based on soil analyses, climate data and performance indicators.
- Planned planting schedules, crop rotation and fertilization to increase yield and reduce costs.

Farm Manager

Fazenda Mundo Novo, Darcinópolis, TO, Brazil · 2021 – 2022

- Managed daily maize and pasture production operations for livestock, from planning to execution.
- Led and trained field teams and took part in negotiations, resource management and decision making.

Technical Consultant

SENAR (CNA System), Imperatriz, MA, Brazil · 2020 – 2021

- Provided technical assistance to 30 beef cattle farms in the interior of Maranhão, bringing technologies and best practices to improve productivity, animal health and efficiency.

Undergraduate Research, Sorghum Breeding

Embrapa Maize and Sorghum (Sete Lagoas, MG) and IFTO, Araguatins Campus · 2017 – 2020

- Under the supervision of Prof. Leonardo Correia, worked with the Embrapa Maize and Sorghum breeding program evaluating dual-purpose biomass sorghum, intended for both biomass production and forage, in the far north of Tocantins.

PROJECTS

Beyond Prediction: Quantifying Certainty in Genotype Performance Across Untested Environments

ONGOING

We are developing an approach to quantify the uncertainty of genotype performance predictions in untested environments, going beyond point estimates to support recommendation decisions with greater confidence.

From Multi-Environment Trials to Data-Driven Genotype Selection in Soybean

ONGOING

In partnership with GDM Seeds, we investigate what models that integrate environmental information actually capture of genotypes × environments interaction. Leveraging continental-scale data (5 breeding programs, 155 locations and 1,232 genotypes, from 2021 to 2023) and the broad representativeness of the company's germplasm, present in about 45% of global soybean production and in over 80% of the soybean area in Brazil, we seek to understand how these models represent genotype adaptation across space and what this means for selection in untested environments.

Tropical wheat expansion into the Brazilian Cerrado

COMPLETED

How can we know where tropical wheat will perform well before it is even planted? We developed a pipeline that combines GIS, factor analytic models and 83 environmental covariates to predict genotype performance in untested environments, supporting the UFV wheat breeding program in expanding the crop across the Brazilian Cerrado. Pixel-by-pixel recommendations outyielded the best genotype indicated by conventional selection tools by 9.8% (rainfed) and 13.4% (irrigated). Published in *Crop Science* (2026).

Early selection of tropical wheat populations

COMPLETED

Can we tell early which populations will give rise to the best lines? We evaluated 25 segregating tropical wheat populations in the F_2, F_3 and F_4 generations across six environments, using models that treat generation as a nested effect and combine different genetic and residual (co)variance structures. The best model pinpointed the populations with the greatest potential to produce superior lines, making selection possible from the earliest generations. Published in *Euphytica* (2025).

qgtools: R functions for quantitative genetics

ON GITHUB

An open-source collection of functions I use daily in quantitative genetics analyses: two-stage multi-environment analysis (BLUEs, BLUPs, weights and Cullis heritability), climate and soil covariate assembly, and spatial interpolation via Random Forest.

PUBLICATIONS

Modeling genotype-by-environment interaction for variety recommendation in tropical wheat

Sousa, J. M. A.; Blasques, G. M.; Condé, A. T.; Silva, A. F.; Eides, J. R.; Fronza, R. T. L.; Fernandes, S. B.; Carneiro, P. C. S.; Bhering, L. L.; Nardino, M.; Dias, K. O. G.

Crop Science, 66(3), e70309 (2026) · doi.org/10.1002/csc2.70309

Statistical genetics models with residual and genetic structures enhance the accuracy of selecting wheat populations

Sousa, J. M. A.; Vieira, E. F. T.; Fronza, R. T. L.; Vieira Netto, J. A. F.; Lima, G. W.; Dias, K. O. G.; Nardino, M.

Euphytica, 221(5), 55 (2025) · doi.org/10.1007/s10681-025-03500-z

Uncovering the parameters and genetic gain in selection strategies of wheat breeding populations

Sousa, J. M. A.; Silva, C. M.; Vieira, E. F. T.; Mezzomo, H. C.; Salvador, O. M.; Nardino, M.

Revista Brasileira de Ciências Agrárias (Agrária), 20(1), e3792 (2025) · doi.org/10.5039/agraria.v20i1a3792

EDUCATION

2026 – present — **Sandwich Ph.D. (Visiting Scholar)**
University of Illinois Urbana-Champaign, USA · Advisor: Prof. Alexander Lipka

2024 – present — **Ph.D. in Genetics and Plant Breeding**
Federal University of Viçosa (UFV) · Advisor: Prof. Kaio Olimpio das Graças Dias

2022 – 2024 — **M.Sc. in Genetics and Plant Breeding**
Federal University of Viçosa (UFV) · Advisor: Prof. Maicon Nardino

2017 – 2020 — **B.Sc. in Agronomic Engineering**
Federal Institute of Tocantins (IFTO), Araguatins Campus · Advisor: Prof. Leonardo Correia

LEADERSHIP & MENTORING

2026 – present — **Mentee, Bayer University Mentoring Program (B4U)**
One-on-one mentoring with a Bayer scientist

2023 – 2024 — **Events Coordinator, GenMelhor**
Study Group in Genetics and Plant Breeding, UFV

2023 – 2024 — **Coordinator, InterGen**
Extension group bringing together seven Brazilian universities

AWARDS

1ST PLACE · JUNE 2026

UFV–Bayer Hackathon 2026

Team Oraculum · Idata, UFV · Viçosa, MG, Brazil

The first edition of Bayer's hackathon held outside the United States. Challenge: train models on data from 2001 to 2007 and predict the performance of maize lines in 2008, integrating genomic and environmental information.

1ST PLACE · JULY 2026

3rd GDM Universities Challenge

QGSL Laboratory, Idata, UFV · Cambé, PR, Brazil

Challenge: develop tools and new methodologies for use in a plant breeding program. The award included a visit to GDM's laboratories.

WORKING GENIUS

I took The Table Group's Working Genius assessment, which identifies six types of genius required for any kind of work. My profile is **Loyal Finisher (ET | TE)**: I draw energy from seeing tasks through to completion and from supporting the people around me. I am the one who responds quickly and delivers reliably.

GENIUS

What energizes me

Tenacity
Seeing work through to completion and making sure results happen.

Enablement
Responding to the team's needs and giving the support that moves ideas forward.

COMPETENCY

I do it well, but it does not energize me

Invention
Creating new solutions and ideas when a problem calls for it.

Galvanizing
Rallying people and getting the team moving.

FRUSTRATION

What drains me

Wonder
Spending a long time speculating without a clear path forward.

Discernment
Judging ideas on intuition alone, without data to rely on.

SKILLS

Scale: Basic · Intermediate · Advanced · Expert

HARD SKILLS

METHODS

Mixed models and GxE	**Advanced**
Genomic selection	**Advanced**
Machine learning	**Intermediate**
Bayesian methods	**Intermediate**

TOOLS

R	**Advanced**
ASReml	**Expert**
BGLR	**Intermediate**
Git / GitHub	**Intermediate**

LANGUAGES

Portuguese	**Native**
English	**Intermediate**

SOFT SKILLS

Commitment to delivery (T)	**Advanced**	
Persistence and results focus (T)	**Advanced**	
Collaboration and teamwork (E)	**Advanced**	
Willingness to support colleagues (E)	**Advanced**	
Problem solving (I)	**Intermediate**	
Leadership and team mobilization (G)	**Intermediate**	
Communication	**Advanced**	
Organization and time management	**Intermediate**	
Adaptability	**Advanced**	

The letters show the Working Genius gear related to each skill (see above): (T) Tenacity, (E) Enablement, (I) Invention and (G) Galvanizing.

PROFESSIONAL REFERENCES

Prof. Kaio Olimpio das Graças Dias
Federal University of Viçosa (UFV) · Ph.D. advisor
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Prof. Alexander E. Lipka
University of Illinois Urbana-Champaign, Department of Crop Sciences · Sandwich Ph.D. advisor
alipka@illinois.edu

Prof. Leonardo Lopes Bhering
Federal University of Viçosa (UFV) · Professor in my graduate program in Genetics and Plant Breeding
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Prof. Maicon Nardino
Federal University of Viçosa (UFV), Department of Agronomy · M.Sc. advisor
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